CLAIMS

What is claimed is:

1. A thermal management system for a battery pack, comprising: a plurality of phase-change modules, each disposed between two adjacent cells of the battery pack; a plurality of bidirectional thermoelectric bridges, each thermally coupling a pair of the phase-change modules; a temperature sensor array; and a controller configured to predict a temperature trajectory of each cell and to drive the bridges such that heat flows from a first cell predicted to exceed an upper threshold toward a second cell predicted to fall below a lower threshold.

2. The system of claim 1, wherein each phase-change module comprises a paraffin-graphite composite having a melting point between 38 degrees C and 42 degrees C.

3. The system of claim 2, wherein the composite is encapsulated in an aluminum pouch.

4. The system of claim 1, wherein each thermoelectric bridge comprises bismuth telluride couples between copper spreaders.

5. The system of claim 1, wherein the controller predicts the temperature trajectory over a horizon of at least 60 seconds.

6. The system of claim 1, wherein the controller uses a reduced-order lumped-capacitance model of the pack.

7. The system of claim 6, wherein the controller identifies parameters of the model from sensor data recorded during operation.

8. The system of claim 1, wherein the controller selects bridge currents by model-predictive optimisation.

9. The system of claim 8, wherein the optimisation penalises electrical energy consumed by the bridges.

10. The system of claim 8, wherein the optimisation constrains a latent-heat state of charge of each module.

11. The system of claim 1, wherein, in a cold-start mode, the controller drives the bridges to return heat stored in the modules to the cells.

12. The system of claim 1, wherein the controller limits a charging current when a sensor reading deviates from a model estimate by more than a threshold.

13. The system of claim 1, wherein the cells are solid-state lithium-metal cells.

14. A method of managing temperature in a battery pack, comprising: measuring cell temperatures; predicting a temperature trajectory of each cell; and driving bidirectional thermoelectric bridges between phase-change modules such that heat flows from cells predicted to overheat toward cells predicted to undercool.

15. The method of claim 14, further comprising storing heat in the phase-change modules during charging and returning the stored heat to the cells during a subsequent cold start.

16. The method of claim 14, wherein predicting comprises solving a reduced-order thermal model.

17. The method of claim 14, further comprising substituting a model estimate for a faulty sensor reading.

18. The method of claim 14, wherein the driving minimises a weighted sum of temperature deviation, bridge energy and rate of change of bridge current.

19. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to perform the method of claim 14.

20. A vehicle comprising the thermal management system of claim 1.